

BUS23-3

Financial Diagnosis & Management Control

From profitability decomposition to control-lever design

ROIC $\rightarrow$ free cash flow $\rightarrow$ diagnosis: the analytical engine of the Finance arc.

Albert School — 22 hours

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1 Intermediate management balances (SIG)

The intermediate management balances (*soldes intermédiaires de gestion*, SIG) decompose the income statement into a cascade that shows how revenue becomes net income, isolating the contribution of each stage.

Definition 1 The SIG cascade, computed in order:

- **Production / commercial margin** = sales – cost of goods sold.
- **Value added (VA)** = production – purchases of goods and external services consumed.
- **Gross operating surplus (EBE / EBITDA)** = VA + operating subsidies – personnel costs – taxes and duties.
- **Operating result (EBIT)** = EBE – depreciation and amortization – net operating provisions + other operating income.
- **Current result before tax** = operating result + financial income – financial expenses.
- **Net result** = current result + extraordinary result – income tax – employee profit-sharing.

Each balance reveals a distinct layer of the profitability chain: value added measures wealth created by the firm; EBE measures profitability of operations before depreciation and financing; EBIT measures operating profitability after the cost of using fixed assets; the current result incorporates the financing structure; the net result is what remains for owners.

2 DuPont decomposition

Definition 2 DuPont decomposition expresses return on equity as a product of three drivers:

$$\text{ROE} = \underbrace{\frac{\text{Net income}}{\text{Revenue}}}_{\text{net margin}} \times \underbrace{\frac{\text{Revenue}}{\text{Total assets}}}_{\text{asset turnover}} \times \underbrace{\frac{\text{Total assets}}{\text{Equity}}}_{\text{leverage}}.$$

The decomposition identifies which driver produces the observed ROE:

- a **margin-driven** return comes from a high net margin (pricing power, cost control);
- a **turnover-driven** return comes from high asset efficiency (sales generated per unit of assets);
- a **leverage-driven** return comes from a high assets-to-equity ratio (debt financing).

Strategic implication: returns driven by margin or turnover reflect operating performance and are more durable; returns driven by leverage are more fragile, because leverage amplifies losses as well as gains and depends on continued access to debt.

3 ROIC and ROA

Return on invested capital and return on assets measure operational efficiency independently of the financing mix.

Definition 3

$$\text{ROIC} = \frac{\text{NOPAT}}{\text{Invested capital}}, \quad \text{NOPAT} = \text{EBIT} \times (1 - t),$$

where t is the effective tax rate and invested capital = equity + net financial debt (equivalently, net working capital + net fixed assets).

$$\text{ROA} = \frac{\text{Net income}}{\text{Total assets}}.$$

ROIC uses NOPAT — operating profit after tax but *before* interest — so it is unaffected by how the firm is financed. This makes it more reliable than ROE when comparing companies with different capital structures: two firms with identical operations but different debt levels have the same ROIC but different ROE. Value is created when ROIC exceeds the weighted average cost of capital.

4 Advanced ratio analysis and WCR drivers

Definition 4 Working Capital Requirement drivers, expressed in days:

$$\text{inventory days} = \frac{\text{inventory}}{\text{COGS}} \times 365,$$

$$\text{receivable days} = \frac{\text{accounts receivable}}{\text{revenue}} \times 365,$$

$$\text{payable days} = \frac{\text{accounts payable}}{\text{purchases}} \times 365.$$

The cash conversion cycle = inventory days + receivable days − payable days.

Definition 5 Interest coverage ratio:

$$\text{interest coverage} = \frac{\text{EBIT}}{\text{interest expense}},$$

measuring how many times operating profit covers interest due.

Modelling the cash impact: reducing the cash conversion cycle by d days releases cash of approximately $d \times$ daily revenue (using the relevant base per driver). Trade-offs: cutting inventory days risks stock-outs and lost sales; cutting receivable days (tighter credit terms) may lose customers; extending payable days strains supplier relationships and may forfeit early-payment discounts.

5 Free cash flow

Free cash flow — not net income — is the cash available to capital providers, and is the quantity valuation uses.

Definition 6 Free cash flow to the firm:

$$\text{FCFF} = \text{NOPAT} + \text{D\&A} - \text{capex} - \Delta \text{WCR}.$$

Free cash flow to equity:

$$\text{FCFE} = \text{FCFF} - \text{interest} \times (1 - t) + \text{net borrowing}.$$

FCFF is the cash generated by operations available to all providers of capital (debt and equity) before financing flows; FCFE is what remains for equity holders after debt service and new borrowing. Net income differs from free cash flow because it includes non-cash charges (D&A), excludes capital expenditure and working-capital investment, and is affected by accruals; only cash can be distributed or reinvested.

6 Accounting adjustments for comparability

Reported financials embed accounting choices that distort comparison; adjustments restore comparability.

Definition 7 Core adjustments:

- **R&D capitalization:** treat R&D as an asset amortized over its useful life rather than expensing it immediately; add back R&D to operating profit and create an R&D asset on the balance sheet.
- **Operating lease treatment:** capitalize leases by recognizing a right-of-use asset and a lease liability; the rent expense becomes depreciation plus interest.
- **Stock-based compensation normalization:** treat stock-based compensation as a real (cash-equivalent) operating expense and adjust share counts for dilution.

Unadjusted comparisons mislead because two firms making different choices — one expensing R&D, one capitalizing; one leasing, one owning; one paying in stock, one in cash — show different margins, asset bases, and leverage despite identical economics. Adjusting puts them on a common basis before ratios are compared.

7 Peer group analysis

Definition 8 A peer group is a set of comparable companies used as a benchmark, selected to match the subject company on business model, then controlled for differences in size, geography, and capital structure.

Procedure: (1) select comparables operating in the same business; (2) compute the same ratios across the group; (3) control for size, geography, and business-model differences; (4) interpret gaps. A ratio gap signals a **genuine** performance difference when peers are truly comparable and the gap persists after controlling for structural factors; it signals a **structural** difference when it is explained by size, geography, accounting regime, or business model rather than performance.

8 Company lifecycle and financial profiles

Definition 9 Diagnostic expectations differ by lifecycle stage:

- **Startup:** negative or low margins, negative free cash flow, high growth, reliance on external financing.
- **Growth:** improving margins, rising revenue, free cash flow often still negative due to reinvestment and WCR growth.
- **Maturity:** stable margins, positive free cash flow, low growth, returns of capital.
- **Decline:** falling revenue and margins, positive but shrinking free cash flow, asset disposals.

The same ratio carries different meaning by stage: negative free cash flow is normal for a growth firm investing ahead of revenue but a warning sign for a mature firm. Diagnostic thresholds must therefore be set relative to the company's lifecycle stage.

9 Budgeting and variance analysis

Definition 10 A **driver-based budget** projects financial outcomes from operational drivers (volume, price, unit cost) rather than from prior-period figures. A **fixed annual budget** is set once for the year; a **rolling forecast** is updated each period to maintain a constant forward horizon.

Definition 11 Variance decomposition compares actual to budget:

- **Volume variance** = (actual quantity – budgeted quantity) × budgeted margin (or price).
- **Price variance** = (actual price – budgeted price) × actual quantity.
- **Mix variance** = effect of a change in the proportion of products sold, holding total volume constant.
- **Standard-vs-actual cost variance** = (standard cost – actual cost) × actual quantity.

A variance is **actionable / operational** when it stems from execution within the unit's control (e.g. an unfavourable cost variance from waste); it is **structural** when it reflects conditions requiring strategic redesign rather than operational correction (e.g. a permanent shift in input prices or demand).

10 Control-lever design and comprehensive diagnosis

Definition 12 A **control-lever system** for a business unit specifies the piloting indicators to monitor, the frequency of monitoring, and the alert thresholds that trigger action.

Design choices: select a small set of indicators tied to the unit's value drivers; set the monitoring frequency to match how fast each indicator moves and how quickly action is needed; set alert thresholds at the level beyond which a deviation requires intervention. A **rolling forecast** outperforms a **fixed annual budget** when the unit's operating conditions are volatile or fast-changing, so that a once-a-year target becomes obsolete; a fixed budget suffices when conditions are stable and predictable.

10.1 Comprehensive financial diagnosis

A comprehensive diagnosis of a real, possibly distressed company synthesizes the prior tools into a single report:

1. compute the SIG cascade to locate where the profitability chain breaks;
2. apply DuPont to identify the failing return driver (margin, turnover, or leverage);
3. compute ROIC to judge operating performance against the cost of capital;
4. analyse WCR drivers to find cash trapped in the operating cycle;
5. compute free cash flow to assess cash generation and solvency;
6. compare against a peer group, with accounting adjustments, to separate genuine from structural underperformance.

The report identifies the failing driver(s), assesses the severity of the situation, and proposes quantified remediation strategies targeted at the diagnosed cause.